

connection to get the software. And in India, this is still a common issue. It is only appropriate then, that an Indian participant in this year's Google Summer Of Code, Shishir Goel, did the work to enable this particular feature in PackageKit. There are many command line utilities that can assist a user in this task, but they are often distribution dependent and cumbersome. PackageKit offers a very easy alternative in the form of service packs. Service packs are merely software packages and its dependencies wrapped in a standard tarball format. The user can select the dependencies to be packed using an additional option. Along with the dependencies, a service pack has a file named *metadata.conf*, which contains the distribution name, version and the date the pack was created.

For the command line junkies, *pkggenpack* is the command client that uses PackageKit to do the work. A simple example would be:

```
# pkggenpack --output=/media/disk/Rahul --package=xpdf
```

This generates a file */media/disk/Rahul/xpdf-fedora-10-i686.servicepack* on my USB key. A friend of mine can take my USB key, go home, insert the USB key and double click on the service pack file to be prompted to install *xdf* along with dependencies included within the service pack itself. That's just one example. You can do much more, including transferring updates. Do refer to the very well-documented man page for more details.

Future plans

This write up is based on the latest stable version of PackageKit and is already a bit outdated at the time of writing, since PackageKit is under very active development. It has a number of upcoming features planned or already available in the development versions.

The on-demand installation feature, like the one I hinted at earlier, is being expanded to cover more than just codecs.

Font support

Let's suppose your native tongue is Hindi. If someone sends you a document in Hindi and you don't have a Hindi font installed on your system, you would normally get an odd mix of characters that makes no sense at all. With newer versions of PackageKit, however, when an application opens a document, which requires a particular type of font that is not installed but available in the distribution's software repository, PackageKit will automatically be able to search (Figure 9) and install it. So if you always wanted to read Greek love letters, here is your chance!

Mime support

PackageKit goes even further. What if you don't have an application installed that can open the document in question? No worries! PackageKit can take care of that as well.

In Figure 9, PackageKit is asking to search for an

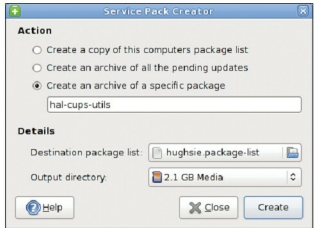


Figure 8: Service Pack Creator to upgrade packages offline later



Figure 9: PackageKit prompts to search for missing fonts

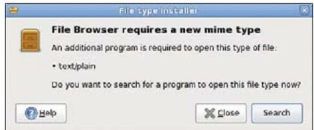


Figure 10: PackageKit prompts to look for an app to access missing mime type

appropriate text editor to open the document, but other documents or mime-types are supported. When multiple applications are able to handle a particular document type, you will be presented a list of choices to pick from with the distribution default shown first.

As you can see, PackageKit aims to enable users to move away from installing software just because they might need it some day, to installing software on-demand instead. There are many other new features in the pipeline as well, but time's running out just now. We'll talk about these another day, maybe. Till then, enjoy PackageKit and join us for questions in order to hack PackageKit at www.packagekit.org.

By: Rahul Sundaram

The author is a Red Hat engineer and active contributor to the Fedora Project, and has contributed a bit to PackageKit as well. He likes to dabble in and write about new and interesting things in the free software space. He can be reached via e-mail at sundaram@fedoraproject.org and via IRC at [mether@freenode.net](https://freenode.net)